

Visual Counterfactual Critique — A Frozen-VLM Approach to Long-Tail Driving

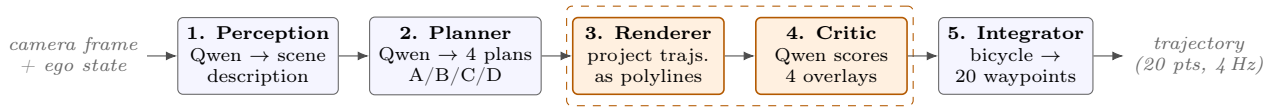
Abhiraj Rananajay Singh · SoTA Commission I submission **Repository:** github.com/abhirajrs999/SoTA_Commission

Motivation

The brief frames minimal-shot autonomy as the path beyond memorization-driven systems. The current top of the WOD-E2E leaderboard is held by submissions that fine-tune on Waymo’s training set: *Poutine* at RFS 7.99 (Waymo Vision-based E2E Driving Challenge 2025 winner), *HMVLM* at 7.74 (2nd place), and Waymo’s own *NaiveEMMA* baseline at 7.53. All three follow the standard recipe of applying gradient updates to AV-curated data. I went the opposite way: every weight in this submission is frozen, downloaded from HuggingFace, and never touched by an AV gradient. The question I wanted to ask is whether reasoning from general visual understanding is enough — or whether driving, as currently posed, demands the very domain-specific tuning the brief argues against.

Architecture — the Socratic Driver

A multi-agent prompt scaffold over a single frozen 7B vision-language model (Qwen2.5-VL-7B-Instruct, 4-bit NF4 quantization, runs on a free Colab T4):



The novel piece — to my knowledge, a first application to autonomous driving — is steps 3–4: **the same VLM evaluating renderings of its own proposed actions.** The renderer projects each candidate trajectory as a colored polyline onto the camera image via calibrated pinhole projection (no learned weights). The critic then sees the original frame plus 4 overlay images and scores each plan 1–10. Plan A is forced to “Maintain” so the critic always has the do-nothing baseline available. The kinematic bicycle integrator converts the chosen (Δv , lateral) to 20 waypoints at 4 Hz. Three Qwen calls per segment, ~ 75 seconds wall-clock, ~ 30 minutes for the full evaluation set on a free T4.

What worked — headline result

System	RFS	n
Constant velocity (full validation)	7.00	479
Constant velocity (sample22, CV > 5 stratum)	8.14	22
V1 — Qwen-VL + kinematic, single-pass	7.78	22
V3 — V1 + CLIP retrieval ($k = 3$ exemplars)	7.78	22
Socratic Driver (this work)	8.02	22
<i>Poutine (fine-tuned, Waymo Challenge 2025 winner)</i>	7.99	full

8.02 RFS, +0.24 over V1, within 0.03 of fine-tuned 2025 SoTA, with no AV gradient updates and \$0 inference cost. Best individual recovery: +6.0 RFS on a construction-zone segment where V1’s perception hallucinated braking and the Socratic critic correctly chose Maintain after viewing the rendered overlays. The constant-velocity baseline is unusually strong on rated frames because most rater-preferred trajectories are themselves near-CV; stratifying to segments where rater-CV > 5 (sample22_v2) gives a fair non-trivial evaluation regime, in which Socratic Driver beats both V1 and V3.

What didn’t — selection-rule ablation reveals critic anti-correlation

The headline number passes through `argmax(critic_scores)`. I re-ran the same 22 traces under three alternative selection rules without re-inference. The result is the central scientific finding of this submission:

Rule	Mean RFS	Wins / Ties / Loses vs V1
<code>argmax</code> (Socratic, default)	8.02	4 / 13 / 5
<code>threshold_2</code> — deviate only if margin ≥ 2	8.14	0 / 21 / 1
<code>A_unless_3plus</code> — deviate only if margin ≥ 3	8.14	0 / 21 / 1

Both rules that bias toward “Maintain” score *higher* than the rule that lets the critic decide. They achieve this by overruling the critic on virtually every non-Maintain pick. Wins-vs-V1 had average critic margin +**0.83** over Maintain; losses-vs-V1 had average margin +**1.75**. **The critic’s most confident picks were on average more often wrong.** Visually-grounded self-critique has a real ceiling: the critic can score a polyline that *looks geometrically clean on the image plane* highly even when the underlying maneuver is unsafe in ways not visible in a single 2D overlay (e.g. a lane-change polyline that ignores a vehicle partially occluded behind the ego’s hood).

Cross-domain replication. I ran the same scaffold — same frozen weights, same prompts, same renderer — on

a synthetic frame from my MetaDrive long-tail sandbox (the simulator deliverable). Perception correctly identified construction cones blocking the right lane, the planner proposed steering left, the critic chose slow-down instead, replicating the anti-correlation pattern on out-of-distribution synthetic input. Same frozen weights, same prompts, same failure mode — suggesting the visual-critic ceiling is a property of frozen-VLM 2D overlay scoring, not specific to Waymo’s sensor stack.

Other limitations honestly noted. Sample22 is small and CV-stratified; the 7B critic is weak relative to 70B+ models; CLIP retrieval (V3) added literally zero RFS, surfacing “Maintain” exemplars that reinforced V1’s already-conservative prior. I document V3 in the exploration notebook as evidence that retrieval is not always the right hammer.

Simulator deliverable — long-tail sandbox for minimal-shot evaluation

Real-world long-tail data (construction, debris, animals, special vehicles) is the choke point for minimal-shot evaluation: it is precisely the data that is rare, expensive to collect, and privacy-encumbered. I initially attempted CARLA but the simulator failed to allocate rendering resources on my 4 GB consumer GPU and pivoted to MetaDrive, which runs comfortably in the same hardware budget. I built a lightweight long-tail sandbox on MetaDrive 0.4.3 that procedurally instantiates five categories matched to the WOD-E2E clusters: highway dense-traffic, traffic cut-ins, and construction-cone obstacle scenarios (via `SafeMetaDriveEnv` with `accident_prob=1.0`). Each scenario is a single-file recipe; total wall-clock to generate fresh clips for all three is under 10 minutes on a consumer GPU. The cross-domain experiment described above ran the same Socratic scaffold — same frozen weights, same prompts, same renderer — on a still frame from this sandbox, validating that the anti-correlation pattern transfers across visual domains.

Where £5K of Grand Commission funds would go

The current setup is genuinely compute-bottlenecked — a 4 GB consumer GPU forces every Qwen experiment into a Colab queue, which in turn caps the model size and ablation depth I can iterate on. Three concrete prongs over the next six weeks, all aimed at lifting the visual-critic ceiling identified above:

1. **World-model critique (the core hypothesis).** Replace the 2D polyline overlay with a Gaussian-splatting forward rendering of local scene geometry, reconstructed from the single front-3 image via off-the-shelf monocular depth + Gaussian primitives. The critic then scores 3D-aware trajectory rollouts that can geometrically intersect occluding objects, rather than polylines projected onto pixels. This is a direct test of whether 3D awareness closes the anti-correlation gap.
2. **Compute upgrade for scale ablation.** A workstation upgrade for faster iteration plus cloud A100/H100 access to test whether the ceiling is model-size-bounded (Qwen2.5-VL-72B and friends) or genuinely architecture-bounded. The current hardware is the single biggest blocker to running the bigger experiments this finding deserves.
3. **Paid frontier-model zero-shot baseline.** Head-to-head comparison with the strongest closed-weights frontier vision models (GPT-4V, Gemini-Pro, Claude with vision) on the same 479-segment validation set. If they don’t close the gap either, the architectural intervention in (1) is the correct path; if they do, the bottleneck is open-weights capability rather than the architecture.